

GroupTrack V2.4

Quickstart Guide

Step-by-step setup for new riders

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What You Need

- Android phone or tablet (tested devices recommended)
- T1000-E Meshtastic radio (one per rider)
- GroupTrack app installed from Google Play Store
- Internet connection for initial map downloads

CRITICAL: Your T1000-E radio MUST remain on firmware 2.6.11. Do NOT accept firmware update prompts. Newer firmware destabilizes BLE and radio config.

CRITICAL: Keep your T1000-E in your cart during rides. Do NOT remove it. Bluetooth disconnects if the radio exceeds 50ft BLE range from your phone.

Step 1: Install GroupTrack

- Open Google Play Store on your Android device
- Search for "GroupTrack"
- Tap Install
- When prompted, grant Location permission (required for GPS tracking)
- When prompted, grant Bluetooth permission (required for radio connection)
- When prompted, grant Notification permission (required for download progress)

If upgrading from a previous version, your downloaded maps and settings are preserved.

Step 2: Power On Your T1000-E Radio

- Press and hold the button on your T1000-E for 3 seconds
- The LED will flash to indicate power on
- Place the radio in your cart, within 50 feet of your phone
- The radio will automatically advertise via Bluetooth

WARNING: Do NOT update the T1000-E firmware. It must stay on version 2.6.11. Newer versions cause BLE disconnection issues.

Step 3: Connect Radio to GroupTrack

- Open GroupTrack
- Tap the Bluetooth icon in the top bar
- Your T1000-E will appear in the device list
- Tap your T1000-E to connect
- Wait for "Connected" status (may take 10-15 seconds)
- The connection indicator turns green when connected

GroupTrack will automatically reconnect to your last radio on future launches.

Step 4: Program Your Radio for the Convoy

Each convoy ride uses a shared radio channel. The ride organizer provides the channel configuration.

- Tap the menu icon (hamburger) in GroupTrack
- Select "Apply Settings" from the menu
- The organizer will share a channel link or QR code
- Scan the QR code or open the link to load channel settings
- Review the settings: channel name, encryption key, radio region
- Tap "Apply to Radio" to program your T1000-E
- Wait for confirmation: "Settings Applied Successfully"

All riders in the convoy must be on the same channel. Verify with your ride organizer before the ride.

Step 5: Download Offline Maps

Download map tiles before your ride so you have offline maps in areas without cell service.

- From the convoy map, tap PLANNING MAP in the top bar
- In the search field at the top, type your ride area (e.g., "St George UT")
- Tap GO to fly to that area
- Tap the DOWNLOAD button in the toolbar
- Adjust the zoom level slider (higher = more detail, larger download)
- Tap "DOWNLOAD REGION" and draw a rectangle around your ride area
- Confirm the download. It will start automatically.
- You can close GroupTrack -- the download continues in the background
- A notification in your system tray shows download progress

Download maps on WiFi before your ride. A typical ride area (30x30 miles) takes 5-15 minutes to download depending on zoom level.

Step 6: Verify Your Downloaded Maps

- After the download completes, tap the DISPLAY pill on the map
- Toggle "Downloaded Areas" ON
- Blue shaded areas show where you have offline tiles
- Verify your ride area is fully covered in blue
- Switch to LOCAL mode using the NET/LOCAL toggle in the top bar
- Confirm the map renders correctly in offline mode
- Switch back to NET mode when done verifying

You can download multiple areas. Each download enters the queue and processes automatically (max 2 at a time).

Step 7: Start Your Ride

- On ride day, open GroupTrack
- Verify your T1000-E is connected (green indicator)
- The convoy map shows your position and all connected riders
- Switch to LOCAL mode if you are out of cell service
- Tap SAT, TOPO, or TOPO+ to choose your preferred tile source
- The map auto-centers on your GPS position

During the ride:

- Your position updates in real-time via GPS
- Other riders appear on the map via Meshtastic mesh radio
- The convoy tracks relative positions of all riders
- Lost rider alerts appear if a rider drops off the mesh

Step 8: Use Display Layers During the Ride

- Tap the floating DISPLAY pill on the convoy map
- Toggle Tracks ON to see your recorded GPS track
- Toggle Trails ON to see trail data overlaid on the map
- Drag the DISPLAY panel anywhere on the map to keep it out of your way
- Tap DISPLAY again to collapse the panel

Step 9: After Your Ride

- Your GPS track is automatically saved
- View your track by toggling Tracks ON in the DISPLAY panel
- Your downloaded maps remain on the device for future rides
- Power off your T1000-E by pressing and holding the button for 5 seconds

Troubleshooting

Radio won't connect:

- Verify T1000-E is powered on (LED flashing)
- Verify Bluetooth is enabled on your phone
- Verify the radio is within 50 feet of your phone
- Try toggling Bluetooth off and on
- Restart GroupTrack

Map tiles not showing offline:

- Verify you downloaded the correct area (check Downloaded Areas in DISPLAY)
- Verify you switched to LOCAL mode
- Try switching tile sources (SAT / TOPO / TOPO+)

Download is slow or stalling:

- Ensure strong WiFi or cellular connection
- Downloads process 2 at a time -- check the queue panel for status
- Large areas with high zoom levels take longer -- try a smaller area first

Quick Reference

Convoy Map: live ride operations, rider tracking, mesh radio status

Planning Map: download maps, view trails, plan routes (accessible via PLANNING MAP button)

DISPLAY panel: toggle map layers (Tracks, Trails, Downloaded Areas)

NET/LOCAL toggle: switch between online and offline map tiles

SAT/TOPO/TOPO+: switch tile source (satellite, topo, USGS topo)

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